Machine Learning

Oral / Viva Questions and Answers

for the 9 practicals of the External Practical

No.	Practical	Questions
-	General ML questions	12
1	Data Pre-processing Techniques (Student dataset)	13
2	Linear Regression (Age vs Price)	12
3	Gradient Descent (y = (x+3)^2, start x0 = 2)	13
4	Logistic Regression	14
5	Decision Tree Classification	14
6	K-Nearest Neighbors (KNN)	14
7	K-Means Clustering (K = 5)	15
8	Association Rule Learning (Apriori)	16
9	Elbow Method for Optimal Number of Clusters	11
-	Formula sheet, code lines, tips	-

Each practical has: **Theory**, a **Marathi summary** (in English letters), **Questions with answers**, and **what the examiner may ask you to change**. Answers use your real results (Student_1, Iris, Loan, Used Car, StudentsPerformance, Mall, Groceries).

How to answer in the oral

- Start every answer with a one-line definition, then add one example. Keep it short.
- If you do not know, say: 'I am not sure, but I think...' and give your best logic. Do not stay silent.
- Know your dataset: file name, number of rows, columns, target column.
- Know your result: accuracy / R2 / best K and what the graph shows.
- Be ready to run the program again and explain each line.
- If the examiner asks for a change (K value, test size, criterion), change only that one number and re-run.
- Mention that you split the data into train and test, scaled for KNN and K-Means, and evaluated with accuracy and confusion matrix.

General Machine Learning Questions

Q1. What is Machine Learning?

Ans: Machine Learning is a part of AI in which a computer learns patterns from data and then makes predictions or decisions on new data, without being programmed with fixed rules for every case.

Q2. What are the types of Machine Learning?

Ans: 1) Supervised learning (data has a target/label: Linear Regression, Logistic Regression, Decision Tree, KNN). 2) Unsupervised learning (no target: K-Means, Apriori). 3) Reinforcement learning (agent learns by reward and penalty).

Q3. Difference between supervised and unsupervised learning?

Ans: Supervised: the data has input X and a known output y, the model learns the mapping. Unsupervised: only input X is given, the model finds hidden groups or patterns itself.

Q4. Difference between regression and classification?

Ans: Regression predicts a continuous number (price, marks). Classification predicts a category/class (Pass/Fail, flower species, Approved/Rejected).

Q5. What are features and target?

Ans: Features (X) are the input columns used to predict. Target (y, label) is the output column we want to predict.

Q6. Why do we split data into training and testing sets?

Ans: Training data is used to teach the model. Testing data is kept unseen to check how well the model works on new data. This shows whether the model really learned or just memorised.

Q7. What is overfitting and underfitting?

Ans: Overfitting: model performs very well on training data but poorly on test data (memorised, too complex). Underfitting: model performs poorly on both (too simple). Fixes for overfitting: more data, simpler model, limit tree depth, cross-validation.

Q8. What does random_state=42 do?

Ans: It fixes the random number seed, so the train-test split (or clustering start) is the same every time you run the program. 42 is just a convention; any number works.

Q9. Which libraries did you use and why?

Ans: pandas: read CSV and handle tables (DataFrame). numpy: numerical arrays and maths. matplotlib: graphs. scikit-learn (sklearn): ML algorithms, scaling, metrics, train_test_split.

Q10. What is the usual ML workflow?

Ans: Load data -> pre-process (clean, encode, scale) -> split into train/test -> choose model -> fit (train) -> predict -> evaluate -> improve.

Q11. What does fit(), predict() and transform() do?

Ans: fit(): the model learns from training data. predict(): gives output for new data. transform(): applies a learned change (like scaling) to data. fit_transform(): does fit and transform in one step.

Q12. What is a DataFrame?

Ans: A pandas table with rows and columns, like an Excel sheet. df.head() shows the first 5 rows, df.info() shows column types, df.describe() shows statistics.

Practical 1: Data Pre-processing Techniques (Student dataset)

Theory

- Data pre-processing = cleaning raw data and converting it into a form that a model can use.
- Steps: check data -> handle missing values -> remove duplicates -> handle outliers -> encode text -> scale features -> train-test split.
- Missing values: drop rows, or fill with mean / median / mode. Use median when outliers exist.
- Outlier (IQR method): Q1 - 1.5*IQR is the lower limit, Q3 + 1.5*IQR is the upper limit, where IQR = Q3 - Q1.
- Encoding: Label Encoding (Male=1, Female=0) and One-Hot Encoding (separate 0/1 column for each category).
- Scaling: Standardization z = (x - mean) / std. Min-Max scaling = (x - min) / (max - min), range 0 to 1.

Marathi summary (sopya bhashet)

Data pre-processing mhanje raw data clean karne. Missing values bharne, duplicate rows kadhne, outlier handle karne, text (Male/Female) la numbers madhe convert karne, scaling karne ani data la train ani test madhe vibhagne. He sagle kelyavar data model sathi ready hoto.

Viva Questions and Answers

Q1. Why is pre-processing needed?

Ans: Real data is dirty: it has missing values, duplicates, outliers, text and different ranges. Models need clean numeric data. If input is bad, output is bad (garbage in, garbage out).

Q2. How did you find missing values?

Ans: df.isnull().sum() counts missing (NaN) values in each column. In Student_1.csv Age has 1, Attendance 2, Marks 1, Income 1 missing.

Q3. How did you fill missing values? Mean or median?

Ans: Age, Attendance, Marks with mean. Income with median, because Income has a big outlier (800000) which would pull the mean up. Median is not affected by outliers.

Q4. What is an outlier? How did you detect it?

Ans: An outlier is a value far away from the other values. I used the IQR method: values below Q1 - 1.5*IQR or above Q3 + 1.5*IQR are outliers. I clipped them to the limit using clip().

Q5. Why did you drop Student_ID and Name?

Ans: They are only identifiers. They have no useful pattern for prediction and can confuse the model.

Q6. Label Encoding vs One-Hot Encoding?

Ans: Label Encoding gives each category a number (Gender: Female=0, Male=1). One-Hot Encoding creates a separate 0/1 column per category (Department_IT, Department_CSE). Use one-hot when categories have no order, so the model does not think 2 is bigger than 1.

Q7. What is drop_first=True in get_dummies?

Ans: It removes the first dummy column to avoid the dummy variable trap (one column can be predicted from the others, which causes redundancy).

Q8. Why do we scale features?

Ans: Columns have different ranges (Age 20, Income 50000). Large-valued columns can dominate distance-based or gradient-based models. Scaling puts all columns on a similar range.

Q9. StandardScaler vs MinMaxScaler?

Ans: StandardScaler makes mean 0 and standard deviation 1. MinMaxScaler squeezes values into 0 to 1. Standard scaling is better when outliers or normal-like data; Min-Max when you need a fixed range.

Q10. What is fit_transform vs transform?

Ans: fit_transform learns the mean/std (or min/max) from the data and applies it. transform only applies already-learned values. For correct practice, fit the scaler on training data only and just transform the test data, to avoid data leakage.

Q11. Did you scale before or after split?

Ans: In this practical I scaled before the split for simplicity. The better method is to split first, fit the scaler on the training set only, then transform both sets. Mention this if asked: it avoids data leakage.

Q12. What does train_test_split(test_size=0.2) mean?

Ans: 20% of rows go to testing and 80% to training. With 30 students: 24 train and 6 test.

Q13. Did the dataset have duplicates?

Ans: df.duplicated().sum() showed 0 duplicates. If there were any, drop_duplicates() removes them.

If the examiner says 'change this and run again'

- Fill with median/mode instead of mean: change .mean() to .median() or .mode()[0].
- Use only MinMaxScaler or only StandardScaler: keep one of the two scaling lines.
- Change split ratio: test_size=0.3 means 70% train, 30% test.

Practical 2: Linear Regression (Age vs Price)

Theory

- Linear Regression predicts a number using a straight line: y = m*x + c (m = slope, c = intercept).
- The best line is found by the least squares method: minimise the sum of squared differences between actual and predicted values.
- Slope m = sum((x - x_mean)*(y - y_mean)) / sum((x - x_mean)^2).
- MAE = average of |error|. MSE = average of error^2. RMSE = square root of MSE. R2 = 1 - SS_res / SS_total (1 = perfect fit).
- Our result: slope is about -4.14 (price falls about 4.14 for each extra year), intercept about 103.6, R2 about 0.99.

Marathi summary (sopya bhashet)

Linear regression mhanje ek sarali rekha (y = mx + c) jyane number predict karto. Ithe Age vadhli ki Price kami hote, mhanun slope negative ahe. R2 jevha 1 chya javal aso tevha model changla.

Viva Questions and Answers

Q1. What is linear regression?

Ans: A supervised algorithm that finds the best straight line y = mx + c between input x and a numeric output y, and uses it to predict new values.

Q2. What is slope and intercept?

Ans: Slope (m, model.coef_) is the change in y for one unit change in x. Intercept (c, model.intercept_) is the value of y when x = 0.

Q3. Why is the slope negative here?

Ans: As the Age of the item increases its Price decreases, so the line goes downward. Slope is about -4.14.

Q4. How does the model find the best line?

Ans: By minimising the cost function, the sum (or mean) of squared errors between actual and predicted y. This is the least squares method.

Q5. What is R2 score? What does 0.99 mean?

Ans: R2 shows how much of the variation in y is explained by x. It lies between 0 and 1 for a good model. 0.99 means the line explains about 99% of the variation: a very good fit.

Q6. Difference between MAE, MSE and RMSE?

Ans: MAE is the average absolute error. MSE is the average squared error (punishes big errors more). RMSE is the square root of MSE, so it is in the same unit as Price.

Q7. Why do we write X = df[['Age']] with double brackets?

Ans: sklearn needs X as a 2D table (rows x columns). Double brackets give a DataFrame (2D). Single brackets give a 1D Series and cause an error.

Q8. What is the difference between simple and multiple linear regression?

Ans: Simple uses one input variable (Age). Multiple uses two or more (Age, Mileage, Brand ...): y = b0 + b1*x1 + b2*x2 + ...

Q9. What is a residual?

Ans: Residual = actual value - predicted value. A good model has small residuals.

Q10. How do you predict the price of a 21-year-old item?

Ans: model.predict(pd.DataFrame({'Age':[21]})). Using the line: 103.6 - 4.14*21 = about 16.7.

Q11. What are the assumptions of linear regression?

Ans: Linear relation between x and y, errors independent, constant variance of errors, errors roughly normal, and no strong multicollinearity (for multiple regression).

Q12. What are the limitations?

Ans: Only models linear relations, sensitive to outliers, and can predict impossible values (like negative price) far outside the data range.

If the examiner says 'change this and run again'

- Predict for another age: model.predict(pd.DataFrame({'Age':[25]})).
- Use 30% test data: test_size=0.3.
- Print only the equation: print('y =', model.coef_[0], '* x +', model.intercept_).

Practical 3: Gradient Descent (y = (x+3)^2, start x0 = 2)

Theory

- Gradient Descent is an optimisation algorithm that finds the minimum of a (cost) function by taking small steps downhill.
- Update rule: x_new = x_old - learning_rate * f'(x). The derivative f'(x) gives the slope.
- For y = (x+3)^2, f'(x) = 2(x+3). Minimum where slope = 0, so x = -3 and y = 0.
- By hand with learning rate 0.1: x0 = 2, slope = 10, x1 = 2 - 0.1*10 = 1; slope = 8, x2 = 1 - 0.8 = 0.2 ... each step the distance to -3 becomes 0.8 times.
- Learning rate too small = very slow. Too large = overshoots and may never settle.

Marathi summary (sopya bhashet)

Gradient descent mhanje donger utarnyasarkhe, lahan lahan paule taakat minimum point shodhne. Ithe x = 2 pasun suru karun, rate 0.1 ne, x = -3 la pohchto, jithe y = 0 hote.

Viva Questions and Answers

Q1. What is gradient descent?

Ans: An iterative algorithm that moves towards the minimum of a function by repeatedly stepping in the opposite direction of the gradient (slope).

Q2. What is the update formula?

Ans: x_new = x_old - learning_rate * derivative. Here derivative = 2*(x+3).

Q3. What is the learning rate?

Ans: The size of each step. Small value = slow but safe. Large value = fast but may overshoot or diverge. We used 0.1.

Q4. What is a cost function?

Ans: A function that measures how wrong the model is (like squared error). Training means minimising it. Here the function y = (x+3)^2 acts as the cost function.

Q5. Why is the derivative of (x+3)^2 equal to 2(x+3)?

Ans: Power rule: d/dx of u^2 is 2u times du/dx, and here du/dx = 1, so it is 2(x+3).

Q6. What is the minimum value and where is it?

Ans: Minimum at x = -3 where y = (-3+3)^2 = 0. Our program prints x as about -3 and y as about 0.

Q7. Show the first step by hand.

Ans: x0 = 2. Slope = 2*(2+3) = 10. x1 = 2 - 0.1*10 = 1. Next slope = 2*(1+3) = 8, x2 = 1 - 0.8 = 0.2.

Q8. What happens if learning rate = 1?

Ans: x_new = x - 2(x+3) = -x - 6. Starting at 2 it jumps to -8, then back to 2, forever: it oscillates and never reaches -3.

Q9. What happens if learning rate is greater than 1 (like 1.1)?

Ans: Each step overshoots more than the last; x moves away from -3 and the value explodes (divergence).

Q10. When does the loop stop?

Ans: After a fixed number of iterations (100) or when the change in x becomes very small (convergence).

Q11. Local vs global minimum?

Ans: Local minimum: lowest point in a small region. Global minimum: lowest point of the whole function. y = (x+3)^2 is a convex (bowl-shaped) function, so it has only one minimum, which is global.

Q12. Types of gradient descent?

Ans: Batch (uses all data per step), Stochastic (one sample per step), Mini-batch (small groups per step).

Q13. Where is gradient descent used?

Ans: Training linear/logistic regression and neural networks: it adjusts weights to minimise the loss.

If the examiner says 'change this and run again'

- Change starting point: x = 5 (the answer stays x = -3).
- Change learning rate: lr = 0.01 (slower, needs more iterations) or lr = 0.5 (reaches -3 in one or two steps).
- Change iterations: range(50).

Practical 4: Logistic Regression

Theory

- Logistic Regression is a CLASSIFICATION algorithm (despite the word 'regression').
- It computes z = b0 + b1*x1 + ..., then passes it through the sigmoid function: sigmoid(z) = 1 / (1 + e^(-z)). Output is a probability between 0 and 1.
- If probability >= 0.5 the class is 1, otherwise 0 (threshold 0.5). Loss function: log loss.
- Set A uses Iris (3 species, multi-class). Set B uses Study_Hours and Attendance to predict Pass (1) / Fail (0), which is binary.
- Metrics: Accuracy = (TP+TN)/total. Precision = TP/(TP+FP). Recall = TP/(TP+FN). F1 = 2*P*R/(P+R).

Marathi summary (sopya bhashet)

Logistic regression he naav regression ahe pan te classification sathi vaparle jate (Pass/Fail, flower type). Te sigmoid function vaprun 0 te 1 madhli probability dete. 0.5 peksha jast tar class 1, nahi tar class 0.

Viva Questions and Answers

Q1. What is logistic regression?

Ans: A supervised classification algorithm that predicts the probability that an input belongs to a class, using the sigmoid function.

Q2. Why is it called regression if it does classification?

Ans: Because internally it uses a linear regression equation (b0 + b1*x1 ...), and then the sigmoid converts that number into a probability.

Q3. What is the sigmoid function?

Ans: sigmoid(z) = 1 / (1 + e^(-z)). It gives an S-shaped curve whose output is always between 0 and 1.

Q4. What is the threshold?

Ans: The cut-off probability, normally 0.5: above it predict class 1, below it class 0.

Q5. Linear vs Logistic regression?

Ans: Linear predicts a continuous number using a straight line. Logistic predicts a class/probability using the sigmoid curve.

Q6. Binary vs multi-class classification?

Ans: Binary has 2 classes (Pass/Fail). Multi-class has more than 2 (Iris has 3 species). sklearn LogisticRegression handles both.

Q7. What is a confusion matrix?

Ans: A table comparing actual and predicted classes: TP, TN, FP, FN. Diagonal values are correct predictions.

Q8. Define TP, TN, FP, FN.

Ans: TP: predicted positive and actually positive. TN: predicted negative and actually negative. FP: predicted positive but actually negative. FN: predicted negative but actually positive.

Q9. Define accuracy, precision, recall, F1.

Ans: Accuracy = (TP+TN)/all. Precision = TP/(TP+FP), how many predicted positives are correct. Recall = TP/(TP+FN), how many actual positives were found. F1 = harmonic mean of precision and recall.

Q10. What is max_iter=200 in LogisticRegression?

Ans: The maximum number of iterations the solver may use to converge. If it is too small you get a convergence warning.

Q11. What is the loss function of logistic regression?

Ans: Log loss (binary cross-entropy): -[y*log(p) + (1-y)*log(1-p)]. It is minimised during training.

Q12. How do you predict for a new student?

Ans: new = pd.DataFrame({'Study_Hours':[5.0], 'Attendance':[75]}); model.predict(new). Output 1 means Pass, 0 means Fail.

Q13. What is a decision boundary?

Ans: The line (or surface) where the probability is exactly 0.5; it separates the two classes.

Q14. Can accuracy mislead?

Ans: Yes, if classes are imbalanced (for example 95% Pass). Then use precision, recall and F1 as well.

If the examiner says 'change this and run again'

- Change test size: test_size=0.2 or 0.3.
- Change threshold: use model.predict_proba(X_test)[:,1] > 0.6.
- Show probabilities: model.predict_proba(new).

Practical 5: Decision Tree Classification

Theory

- A Decision Tree asks yes/no questions on features and splits the data into branches until it reaches a decision (leaf).
- Parts: root node (first question), internal nodes (questions), branches (answers), leaf nodes (final class).
- Split quality: Gini = 1 - sum(p^2). Entropy = - sum(p * log2(p)). Information Gain = Entropy(parent) - weighted Entropy(children). Lower impurity is better.
- sklearn uses the CART algorithm (binary splits). Default criterion is Gini.
- A very deep tree overfits. Control with max_depth or pruning. Set A uses Iris (gini). Set B uses Loan Approval (entropy, max_depth=4).

Marathi summary (sopya bhashet)

Decision tree mhanje zadasarkhe prashna-uttar (yes/no) vichare tharvaycha. Root pasun leaf paryant jaun class thharte. Gini kinva entropy vaprun changla split nivadla jato. Khup khol zad kela tar overfitting hote, mhanun max_depth thevto.

Viva Questions and Answers

Q1. What is a decision tree?

Ans: A tree-shaped model that splits data using feature conditions and predicts the class at the leaf.

Q2. What are root, internal, branch and leaf nodes?

Ans: Root: top node with the first split. Internal node: a test on a feature. Branch: outcome of a test. Leaf: final prediction (class).

Q3. What is Gini index?

Ans: Gini = 1 - sum(p^2) where p is the proportion of each class in a node. 0 means pure node (only one class). For 2 classes the maximum is 0.5.

Q4. What is entropy?

Ans: Entropy = -sum(p*log2(p)). It measures disorder. 0 means pure; for 2 classes the maximum is 1.

Q5. What is information gain?

Ans: The drop in entropy after a split: Entropy(parent) - weighted average Entropy(children). The split with the highest gain is chosen.

Q6. Gini vs Entropy?

Ans: Both measure impurity and usually give similar trees. Gini is faster (no log). Entropy is based on information theory.

Q7. What is overfitting in a tree? How to prevent it?

Ans: A deep tree memorises training data and fails on new data. Prevent with max_depth, min_samples_split, pruning, or more data.

Q8. What does max_depth=4 mean?

Ans: The tree can have at most 4 levels of questions below the root, which keeps it small and general.

Q9. Why is scaling not needed for decision trees?

Ans: Trees split on thresholds of a single feature at a time, so the feature range does not matter.

Q10. Advantages and disadvantages?

Ans: Advantages: easy to understand and visualise, handles numeric and categorical data, no scaling. Disadvantages: overfits easily, small data changes can change the tree, less accurate than ensembles.

Q11. What did you do with missing values in the loan data?

Ans: Numeric columns (Age, Income, LoanAmount, CreditScore) were filled with median and Gender with mode, because the tree cannot handle NaN.

Q12. How do you see which feature is most important?

Ans: model.feature_importances_ gives the importance of each feature (in our loan tree Credit Score is about 0.70 and Income about 0.26, so these two matter most).

Q13. Decision Tree vs Random Forest?

Ans: Random Forest builds many trees on random samples and takes their majority vote, so it overfits less and is more accurate.

Q14. What is plot_tree used for?

Ans: It draws the trained tree: split condition, gini/entropy, samples and class in every node. filled=True colours nodes by class.

If the examiner says 'change this and run again'

- Switch criterion: criterion='gini' or criterion='entropy'.
- Change depth: max_depth=3 (smaller tree) or remove max_depth (full tree).
- Print tree as text: from sklearn.tree import export_text; print(export_text(model, feature_names=list(X.columns))).

Practical 6: K-Nearest Neighbors (KNN)

Theory

- KNN classifies a new point by looking at its K nearest training points and taking the majority class.
- Distance: Euclidean = sqrt((x1-y1)^2 + (x2-y2)^2 + ...). Manhattan = |x1-y1| + |x2-y2| + ... sklearn default is Euclidean.
- KNN is a lazy learner (no real training; it just stores the data) and non-parametric.
- Scaling is a MUST because distance is used. Small K = overfitting (noisy), large K = underfitting (too smooth). Use odd K to avoid ties.
- Set A uses Used Car data: price is converted into 3 classes (Low, Medium, High). Set B uses Iris (K=5).

Marathi summary (sopya bhashet)

KNN mhanje navin point chya javalche K shejari point pahun, tyanchya pekshi jast kontya class che ahet te baghun class thharvto. Distance vaparle jate mhanun scaling garjecha ahe. K lahan asel tar overfit, motha asel tar underfit.

Viva Questions and Answers

Q1. How does KNN work?

Ans: For a new point it calculates the distance to all training points, picks the K nearest, and gives the class that occurs most among them.

Q2. Why is KNN called a lazy learner?

Ans: It does no training. fit() just stores the data, and all the work (distance calculation) is done at prediction time.

Q3. How do you choose K?

Ans: Try different values (1 to 20), compare test accuracy, and choose the best. Prefer an odd K for binary problems. Default in sklearn is 5.

Q4. What happens for very small or very large K?

Ans: K = 1: very sensitive to noise, overfits. Very large K: ignores local patterns, underfits (in the extreme it predicts the majority class always).

Q5. Why is feature scaling important in KNN?

Ans: Distance is dominated by large-valued features (kms 50000 vs year 2015). Scaling gives all features equal importance.

Q6. What is Euclidean distance?

Ans: Straight-line distance: sqrt(sum((a_i - b_i)^2)). sklearn uses Minkowski with p=2, which equals Euclidean.

Q7. Euclidean vs Manhattan?

Ans: Euclidean is the straight line distance. Manhattan is the sum of absolute differences along each axis (like walking along city blocks).

Q8. Why did you convert the car price into classes?

Ans: KNN classifier needs class labels. Price is a number, so I cut it into 3 equal-sized groups using pd.qcut: 0 = Low, 1 = Medium, 2 = High. (For a numeric price you could use KNeighborsRegressor.)

Q9. How did you clean the car price and kms columns?

Ans: They were text like '4.45 Lakh' and '22,402 km'. I removed commas and used a regular expression (str.extract) to keep only the number. Crore prices were multiplied by 100 to convert to Lakh.

Q10. Why is the car accuracy only about 57%?

Ans: Only kms, fuel type and year were used. Price also depends on brand, model, condition, etc. Adding more features would improve it.

Q11. Advantages and disadvantages of KNN?

Ans: Advantages: simple, no training, works for multi-class. Disadvantages: slow for large data, needs scaling, sensitive to irrelevant features and noise, needs memory to store all data.

Q12. KNN vs K-Means?

Ans: KNN is supervised classification (needs labels, K = number of neighbours). K-Means is unsupervised clustering (no labels, K = number of clusters). The K means different things.

Q13. What is the curse of dimensionality?

Ans: With many features, all points become almost equally far from each other, so distance loses meaning and KNN performs badly.

Q14. Can KNN be used for regression?

Ans: Yes: KNeighborsRegressor predicts the average of the K nearest neighbours' values.

If the examiner says 'change this and run again'

- Change K: n_neighbors=3 or 7.
- Change distance: KNeighborsClassifier(n_neighbors=5, metric='manhattan').
- Remove scaling and compare accuracy to show why scaling matters.

Practical 7: K-Means Clustering (K = 5)

Theory

- K-Means is an UNSUPERVISED clustering algorithm: it groups similar data points without any labels.
- Steps: 1) choose K. 2) pick K random centroids (k-means++ chooses them smartly). 3) assign each point to the nearest centroid. 4) move each centroid to the mean of its points. 5) repeat 3 and 4 until the centroids stop changing.
- Cost = WCSS (inertia) = sum of squared distances of points to their own centroid. Lower is tighter.
- Set A: StudentsPerformance (math, reading, writing scores, scaled). Set B: Mall Customers (Annual Income and Spending Score).
- In the mall data the 5 clusters are customer segments such as high income-high spending, low income-low spending, etc.

Marathi summary (sopya bhashet)

K-Means mhanje label nasleli data che similar goshti che group (cluster) karne. Prathamt K centroids ghyaycha, pratyek point javalchya centroid kade pathvaycha, mag centroid halvaycha. Centroid halane band zale ki cluster tayar.

Viva Questions and Answers

Q1. What is clustering?

Ans: Grouping data points so that points in the same group are similar and points in different groups are different. It is unsupervised.

Q2. Explain the K-Means algorithm.

Ans: Choose K. Place K centroids. Assign every point to its nearest centroid. Recompute each centroid as the mean of its assigned points. Repeat until no point changes cluster.

Q3. What is a centroid?

Ans: The centre (mean position) of all points in a cluster.

Q4. What is WCSS / inertia?

Ans: Within-Cluster Sum of Squares: the total squared distance of every point from its own cluster centroid. Available as model.inertia_.

Q5. Is K-Means supervised or unsupervised?

Ans: Unsupervised. There is no target column; we only give X.

Q6. How did you choose K = 5?

Ans: It is given in the practical. In general K is chosen using the Elbow method (Practical 9) or the silhouette score.

Q7. What is k-means++?

Ans: A smarter way to choose the starting centroids so they are far apart. It gives better and more stable results than purely random starts.

Q8. What is n_init=10?

Ans: K-Means is run 10 times with different starting centroids and the best result (lowest inertia) is kept.

Q9. Why did you scale the Students data but not the Mall data?

Ans: The three score columns are similar in scale, but scaling is a safe habit because K-Means uses distance. In the Mall data both columns (income in k$ and score 1-100) already have comparable ranges, so scaling was skipped.

Q10. What does fit_predict() do?

Ans: It fits K-Means on the data and returns the cluster number (0 to 4) of every row.

Q11. What do the red X marks in the plot show?

Ans: The final centroids (cluster centres).

Q12. Limitations of K-Means?

Ans: K must be chosen in advance. Sensitive to outliers and starting points. Works best for round, similar-sized clusters. Needs numeric data.

Q13. What are the real-life uses?

Ans: Customer segmentation (mall/marketing), grouping students by performance, image compression, document grouping.

Q14. KNN vs K-Means?

Ans: KNN: supervised classification using labelled data. K-Means: unsupervised clustering with no labels.

Q15. What is hard clustering?

Ans: Each point belongs to exactly one cluster (K-Means). In soft clustering a point can belong to many clusters with probabilities.

If the examiner says 'change this and run again'

- Change number of clusters: n_clusters=3 or 4.
- Use other columns: X = df[['Age','Annual Income (k$)']].
- Print cluster centres: print(model.cluster_centers_).

Practical 8: Association Rule Learning (Apriori)

Theory

- Association rule learning finds items that are bought together (market basket analysis), e.g. Bread -> Butter.
- A rule A -> B: A = antecedent (if part), B = consequent (then part).
- Support(A) = transactions containing A / total transactions. Confidence(A->B) = support(A and B) / support(A). Lift(A->B) = confidence / support(B).
- Lift > 1: positive association (bought together more than by chance). Lift = 1: independent. Lift < 1: negative association.
- Apriori principle: every subset of a frequent itemset must also be frequent; so if an itemset is infrequent, all its supersets are infrequent (this prunes the search).
- Set A: T1 to T5 (5 transactions). Set B: Groceries dataset (38,765 rows, 14,963 baskets, 167 items, min support 0.1%).

Marathi summary (sopya bhashet)

Association rule mhanje dukanat konte sathya ekatra vikat ghetle jatat te shodhne (Bread -> Butter). Support mhanje kiti vela aala, confidence mhanje A ghetlyavar B kiti vela ghetla, lift 1 peksha jast asel tar te changla rule.

Viva Questions and Answers

Q1. What is association rule learning?

Ans: An unsupervised method that finds interesting relationships (rules) between items in large transaction data, like 'customers who buy bread also buy butter'.

Q2. What is market basket analysis?

Ans: Analysing what items customers buy together, to plan shelf placement, offers and recommendations.

Q3. Define support, confidence and lift.

Ans: Support = count(A and B)/N. Confidence = support(A and B)/support(A) = P(B given A). Lift = confidence/support(B) = how much more often A and B occur together than expected by chance.

Q4. Calculate support of Bread in T1-T5.

Ans: Bread is in T1, T2, T3, T5 = 4 of 5 transactions, so support = 4/5 = 0.8.

Q5. Calculate Bread -> Butter.

Ans: Bread and Butter together are in T1, T2, T5 = 3/5 = 0.6. Confidence = 0.6/0.8 = 0.75. Support(Butter) = 4/5 = 0.8, so Lift = 0.75/0.8 = 0.94.

Q6. Calculate Jam -> Bread.

Ans: Jam is in T1, T2 (support 0.4) and both also contain Bread, so confidence = 0.4/0.4 = 1.0. Lift = 1.0/0.8 = 1.25, so a positive association.

Q7. What does lift = 1.25 mean?

Ans: Customers who buy Jam are 1.25 times more likely to buy Bread than an average customer.

Q8. What does lift less than 1 mean?

Ans: A negative association: the items occur together less often than if they were independent (like Bread -> Butter, 0.94).

Q9. What is the Apriori principle?

Ans: If an itemset is frequent then all its subsets are frequent; if an itemset is infrequent, all its supersets are infrequent. This lets us skip many candidates.

Q10. Explain the Apriori steps.

Ans: 1) Set min support and min confidence. 2) Find frequent 1-itemsets. 3) Join them to make candidate 2-itemsets, keep the frequent ones, then 3-itemsets and so on. 4) From each frequent itemset generate rules. 5) Keep rules with confidence at least the minimum.

Q11. What are min_support and min_conf in your program?

Ans: Set A: min support 0.4 (40%) and min confidence 0.6. Groceries: min support 0.001 (0.1%) and min confidence 0.1. Groceries needs a low support because there are 167 different items.

Q12. How did you make baskets for the Groceries data?

Ans: Rows are single items, so I grouped by Member_number and Date: all items one customer bought on one date form one basket (14,963 baskets).

Q13. What are antecedent and consequent?

Ans: In A -> B, A is the antecedent (the 'if' items) and B is the consequent (the 'then' items).

Q14. Apriori vs FP-Growth?

Ans: Apriori generates candidate itemsets and scans data many times (slow). FP-Growth builds a compact FP-tree and finds frequent itemsets without candidate generation, so it is faster.

Q15. Is it supervised or unsupervised?

Ans: Unsupervised: there is no target variable.

Q16. Applications?

Ans: Supermarket layout and offers, product recommendation (e-commerce), cross-selling, medical diagnosis patterns, web usage mining.

If the examiner says 'change this and run again'

- Change minimum support or confidence values at the top of the program (min_support, min_conf).
- Show only rules with lift above 1.2: add a condition if lift > 1.2.
- Show only the top 5 rules: rules[:5].

Practical 9: Elbow Method for Optimal Number of Clusters

Theory

- The Elbow method helps to choose K for K-Means.
- Run K-Means for K = 1 to 10 and record the WCSS (inertia) each time. WCSS always decreases as K increases (it becomes 0 when K = number of points).
- Plot K (x-axis) against WCSS (y-axis). The curve drops sharply and then flattens. The point where it bends, like an elbow, is the best K.
- Reason: after the elbow, adding more clusters gives only a small improvement, so extra clusters are not worth it.
- Another method: Silhouette score (range -1 to 1; higher is better).

Marathi summary (sopya bhashet)

Elbow method mhanje K-Means sathi yogya K shodhne. K = 1 pasun 10 paryant WCSS kadhaycha ani graph kadhaycha. Graph jithe kohalya sarkha vakto (bend hoto) toch changla K.

Viva Questions and Answers

Q1. What is the Elbow method and why is it used?

Ans: It is used to find the optimal number of clusters (K) for K-Means by plotting WCSS against K and picking the bend of the curve.

Q2. What is WCSS?

Ans: Within-Cluster Sum of Squares: the sum of squared distances between each point and its cluster centroid. In sklearn it is model.inertia_.

Q3. Why does WCSS decrease when K increases?

Ans: More clusters means every point is closer to its centroid. With K equal to the number of points, WCSS is 0.

Q4. Why not just pick the K with the lowest WCSS?

Ans: That would be K = number of points: useless clusters. We want a small K after which the improvement is small, the elbow.

Q5. How do you read the graph?

Ans: Look for the point after which the line becomes flat. The K at that bend is the optimal K (in our run WCSS falls 3000, 1254, 788, 605 for K=1 to 4 and then only slowly, so the elbow is at about K = 3 or 4).

Q6. What range of K did you use and why?

Ans: K = 1 to 10 using a for loop. A reasonable range for a small dataset.

Q7. What is the loop doing in your program?

Ans: For each K it creates KMeans(n_clusters=k), fits the scaled data, and appends model.inertia_ to the wcss list. Then plt.plot(range(1,11), wcss) draws the curve.

Q8. What if the elbow is not clear?

Ans: Use the silhouette score or the gap statistic, or use domain knowledge.

Q9. What is the silhouette score?

Ans: A measure of how well a point fits its own cluster compared to the nearest other cluster. Ranges from -1 to 1; closer to 1 is better.

Q10. Why did you scale the data first?

Ans: K-Means uses distance, so scaling makes sure no score column dominates.

Q11. What is the relation between Practical 7 and Practical 9?

Ans: Practical 9 finds the best K; Practical 7 then runs K-Means with that K (here K = 5 was given).

If the examiner says 'change this and run again'

- Change range of K: range(1, 15).

- Add silhouette score: from sklearn.metrics import silhouette_score; silhouette_score(X_scaled, model.labels_) for K >= 2.
- Run on Mall data: X = df[['Annual Income (k$)','Spending Score (1-100)']].

Formula Sheet

Name	Formula
Accuracy	(TP + TN) / (TP + TN + FP + FN)
Precision	TP / (TP + FP)
Recall	TP / (TP + FN)
F1 score	2 * Precision * Recall / (Precision + Recall)
Linear regression	y = m*x + c
MSE / RMSE	mean((y - y_pred)^2) / sqrt(MSE)
R2 score	1 - SS_res / SS_total
Sigmoid	1 / (1 + e^(-z))
Gradient descent	x_new = x_old - learning_rate * f'(x)
Gini	1 - sum(p^2)
Entropy	- sum(p * log2(p))
Information gain	Entropy(parent) - weighted Entropy(children)
Euclidean distance	sqrt((x1-y1)^2 + (x2-y2)^2 + ...)
Manhattan distance	|x1-y1| + |x2-y2| + ...
Standardization	z = (x - mean) / std
Min-Max scaling	(x - min) / (max - min)
IQR outlier limits	Q1 - 1.5*IQR and Q3 + 1.5*IQR (IQR = Q3 - Q1)
WCSS	sum of squared distances of points to their cluster centroid
Support	count(A) / N
Confidence (A->B)	support(A and B) / support(A)
Lift (A->B)	confidence / support(B)

Important Code Lines - What They Do

Code	Meaning
`pd.read_csv('file.csv')`	Reads the CSV file into a DataFrame.
`df.head() / df.info() / df.describe()`	First 5 rows / column types and missing counts / statistics.
`df.isnull().sum()`	Counts missing values in each column.
`df['Age'].fillna(df['Age'].mean())`	Replaces missing Age values with the mean.
`LabelEncoder().fit_transform(col)`	Converts text categories to numbers.
`pd.get_dummies(df, columns=[...])`	One-hot encoding.
`StandardScaler().fit_transform(X)`	Scales features to mean 0, std 1.
`train_test_split(X, y, test_size=0.2, random_state=42)`	Splits data into 80% train and 20% test.
`model.fit(X_train, y_train)`	Trains the model.
`model.predict(X_test)`	Predicts for test data.
`accuracy_score(y_test, y_pred)`	Fraction of correct predictions.
`confusion_matrix(y_test, y_pred)`	Table of correct/wrong predictions per class.
`classification_report(y_test, y_pred)`	Precision, recall, F1 for each class.
`model.coef_ / model.intercept_`	Slope and intercept of the regression line.
`KMeans(n_clusters=5, n_init=10)`	K-Means with 5 clusters, run 10 times with different starts.
`model.inertia_`	WCSS of the fitted K-Means.
`plt.scatter() / plt.plot() / plt.show()`	Draw points / draw a line / display the graph.
`pd.qcut(df['price'], 3, labels=[0,1,2])`	Splits price into 3 equal-sized groups.